

Surendra Reddy Medapati

Salt Lake City, UT | Mobile: +1 (813) 570- 3675 | Email: surendrareddy444044@gmail.com | LinkedIn: [LinkedinProfile](#)

EXPERIENCE

AL/ML Engineer | Cypress ILI | United States

Sep 2024 – Present

- Increased anomaly detection accuracy by 30% by analyzing \$2M pipeline inspection data using **Python** libraries such as TensorFlow and Keras, integrating **MLOps** practices for continuous model monitoring and deployment.
- Streamlined **ETL** workflows with **Airflow** and **AWS Glue** and **Redshift**, reducing data processing time by 20% and ensuring efficient data querying for anomaly detection.
- Developed 15+ **Looker** dashboards to track **KPIs**, reducing manual reporting time by 30% and enabling real-time data insights for stakeholders, leading to more informed and timely decision-making.

Data Engineer | Target Arc | United States

Jun 2024 – Aug 2024

- Increased data processing efficiency by 30% by designing Dockerized **ETL** pipelines with **Airflow** and implementing **FastAPI** for real-time data ingestion and **API** integration.
- Extracted and processed 10,000+ biomarkers using fine-tuned **LLMs** (BioBERT, SciBERT), leveraging **AWS SageMaker** to improve biomedical text processing accuracy by 20%.
- Leveraged **GraphQL** to query data and utilized **Python** with **RDFLib** to generate **TTL** files, enhancing data accessibility and enabling seamless integration with **Grafana**.

Data Engineer Intern | Proactive IT | United States

Jan 2024 – May 2024

- Conducted financial data analysis using **Python** and **R**, identifying trends across datasets with 1M+ rows to improve processing workflows.
- Optimized **SQL** databases with indexing and vacuuming, enabling real-time updates in **Power BI** dashboards for multiple projects.
- Designed scalable **AWS** solutions (EC2, Lambda, S3, Redshift), enhancing data accessibility and reducing infrastructure costs.

Data Analyst | Cognizant Technology Solutions | India

Aug 2020 – Jul 2022

- Reduced operational costs by 40% by leading the **ETL migration** of 1,000+ healthcare datasets from **SAS** to **AWS Redshift**, improving data reliability and analytics performance.
- Enhanced query performance by through indexing, partitioning, and optimization of **AWS Redshift SQL** queries, reducing report generation time by 20 minutes per query.
- Boosted data retrieval by 30% through Hadoop-based data lakes using **MapReduce**, **Hive** and **EMR** for real-time analytics.

Data Analyst Intern | BSNL | India

Jan 2020 – May 2020

- Created **Looker** dashboards integrating data from **Excel** and **Oracle SQL**, improving decision-making efficiency by providing real-time insights.
- Migrated 200+ **SQL Server** objects to **Snowflake**, optimizing workflows and enhancing cross-platform reporting capabilities.

PROJECTS

Publication on Recognition of Facial Expression | Python, CNN, Pytorch, Computer Vision, Scikit

February 2024

- Published in Springer for developing a 98%-accurate facial expression detection model using Haar Cascade and CNN. [\[LINK\]](#)

Safe URL Inspector | Python, Machine Learning, Selenium, SSL, Node JS, Tableau, GitHub

January 2024

- Developed an application using Selenium and Regex to extract URLs and classify them with a Random Forest model.

SKILLS

- Programming Languages:** Python, FastAPI, R, SAS, SQL
- Data & Cloud Technologies:** Big Data, AWS, Databricks, Snowflake, SQL, MongoDB, GraphQL
- Visualization & ML Tools:** Tableau, Power BI, DOMO, Looker, CNN, NLP, Keras, TensorFlow, Scikit-learn, Pandas, LLMs
- Platforms & Tools:** Docker, Kubernetes, GitHub, Jira, Shell, Figma, Airflow, A/B testing
- Certifications:** AWS Certified Cloud Practitioner, AWS Certified Developer Associate, DataCamp SQL Associate

EDUCATION

Masters in computer science, Saint Leo University, Tampa, FL

GPA: 3.79

Aug 2022 - May 2024

Bachelors in computer science, SASTRA University, Tamil Nadu, India

GPA: 3.50

Jun 2016 - Sep 2020